TECHNICAL DILIGENCE REVIEW - RECORD 022

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-022

1. Scope of the delivery-process review and its limits

Scope of the report at TDR-2026-28 is the engineering delivery process, pipeline metrics and change-management controls, and the cut-off for evidence is 2026-06-18. 2026-06-18 is the evidence cut-off for TDR-2026-28, whose subject is the engineering delivery process, pipeline metrics and change-management controls. This report relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. One limitation of TDR-2026-28 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. The reader of TDR-2026-28 should treat each conclusion as bounded by the section it appears in.

Method in this report was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. Access for the report at TDR-2026-28 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. The present report closes with 7 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. The open items carried forward from this report are set out below; a reader should treat the committed dates as intentions rather than facts. D. Achterberg reports management's own position wherever it differs from the reviewer's, and does not merge the two.

The reviewer notes that the present report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. Where this report reports a supplier's test result, the reviewer read the certificate and did not witness the test. Interview evidence in the present report came from 9 sessions across 8 teams, each minuted and circulated for correction. The interview programme behind TDR-2026-28 covered 8 teams and produced 8 follow-up requests, of which most were satisfied. D. Achterberg has resisted rounding the findings towards a comfortable conclusion in either direction.

TDR-2026-28 covers 9 of 34 services; the untested remainder is listed so a reader can see what was not reached. For the report at TDR-2026-28 the reviewer sampled 3 sites, including the largest deployment and the newest. Ardent Peak commissioned report TDR-2026-28 from Pinehurst Systems Assurance to test the engineering delivery process, pipeline metrics and change-management controls ahead of the Series B financing. The subject of the present report is the engineering delivery process, pipeline metrics and change-management controls, examined for Ardent Peak by Pinehurst Systems Assurance over 7 weeks ending 2026-06-18. D. Achterberg has separated what the company has built from what the company can demonstrate about what it has built.

The convention throughout report TDR-2026-28 is that an absence of evidence is reported as such rather than as evidence of absence. TDR-2026-28 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Where this report cannot conclude, it says what evidence would let it conclude. D. Achterberg has dated each open item in report TDR-2026-28 so that a reader can see which have been open longest rather than only how many there are. D. Achterberg has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

Interviewees for this report were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. 9 interviews inform the present report. Where an interview is the sole source for a claim, the claim is graded limited. The instruction behind the present report was narrow: examine the engineering delivery process, pipeline metrics and change-management controls and report by 2026-06-18. As instructed, Pinehurst Systems Assurance confined the report at TDR-2026-28 to the engineering delivery process, pipeline metrics and change-management controls; three adjacent questions raised during the work are listed as follow-ups rather than answered here. Where this report uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2026-28)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-22-01	the compensating control was not tested	limited	does not change the conclusion	a document the company has not produced
OQ-22-02	the second bench rig does not cover this path	strong	is reflected in the post-close plan	an independent measurement

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-22-03	the measurement basis changed during the reviewed period	limited	narrows the conclusion to the tested scale	a decision from management
OQ-22-04	the population completeness could not be proven	limited	would change the conclusion if resolved adversely	one further interview
OQ-22-05	the compensating control was not tested	moderate	narrows the conclusion to the tested scale	a witnessed test
OQ-22-06	two teams each believe the other owns this	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-22-07	the remediation landed without a regression test	strong	does not change the conclusion	one further interview
OQ-22-08	the objective is met on the company's own measurement only	strong	narrows the conclusion to the tested scale	a document the company has not produced
OQ-22-09	two teams each believe the other owns this	moderate	narrows the conclusion to the tested scale	one further interview
OQ-22-10	the compensating control was not tested	moderate	would change the conclusion if resolved adversely	a document the company has not produced
OQ-22-11	the objective is met on the company's own measurement only	strong	leaves one section unconcluded	a decision from management
OQ-22-12	two teams each believe the other owns this	strong	would change the conclusion if resolved adversely	one further interview
OQ-22-13	the retention window closed before the raw data was requested	strong	narrows the conclusion to the tested scale	one further interview
OQ-22-14	the population completeness could not be proven	moderate	does not change the conclusion	a witnessed test
OQ-22-15	the population completeness could not be proven	strong	does not change the conclusion	a decision from management
OQ-22-16	the objective is met on the company's own measurement only	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-22-17	the retention window closed before the raw data was requested	moderate	does not change the conclusion	a decision from management
OQ-22-18	the declined item has no expiry date	moderate	does not change the conclusion	one further interview
OQ-22-19	the measurement basis changed during the reviewed period	limited	is reflected in the post-close plan	a re-run of the original test
OQ-22-20	the objective is met on the company's own measurement only	moderate	would change the conclusion if resolved adversely	a re-run of the original test
OQ-22-21	the remediation landed without a regression test	limited	is reflected in the post-close plan	an independent measurement
OQ-22-22	the second bench rig does not cover this path	limited	narrows the conclusion to the tested scale	a re-run of the original test
OQ-22-23	the objective is met on the company's own measurement only	limited	would change the conclusion if resolved adversely	a witnessed test
OQ-22-24	the retention window closed before the raw data was requested	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-22-25	the objective is met on the company's own measurement only	strong	would change the conclusion if resolved adversely	a decision from management
OQ-22-26	the compensating control was not tested	strong	narrows the conclusion to the tested scale	a document the company has not produced
OQ-22-27	the population completeness could not be proven	limited	is reflected in the post-close plan	one further interview
OQ-22-28	two teams each believe the other owns this	strong	narrows the conclusion to the tested scale	a decision from management

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-22-29	two teams each believe the other owns this	moderate	does not change the conclusion	a witnessed test
OQ-22-30	the objective is met on the company's own measurement only	limited	leaves one section unconcluded	a decision from management
OQ-22-31	the population completeness could not be proven	moderate	does not change the conclusion	a re-run of the original test
OQ-22-32	the measurement basis changed during the reviewed period	moderate	is reflected in the post-close plan	a witnessed test

2. Pipeline stages, durations and gate configuration

Simulation suite, short is blocking in the merge path. Simulation suite, short adds little on a clean run and a great deal on a contended one to every change that passes through it. Simulation suite, short was stable in every respect the reviewer could measure, and no one the reviewer asked could account for it. simulation suite, short is reported at the evidence strength available and has not been promoted to fit the narrative.

3 h 40 m is the mean duration of simulation suite, nightly, and 4 h 25 m is the figure the reviewer would plan against. In lead-time terms simulation suite, nightly contributes little on a clean run and a great deal on a contended one. Simulation suite, nightly became slower and more reliable at the same time across the measured window. simulation suite, nightly was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Mean duration of hardware-in-the-loop, bench is 1 h 15 m; the tail sits at 2 h 05 m. The material point on hardware-in-the-loop, bench is that queueing on two rigs is the main contributor to firmware lead time. Over the reviewed period the hardware-in-the-loop, bench stage was stable in every respect the reviewer could measure. hardware-in-the-loop, bench was cross-checked against the engineering record supplied for TDR-2026-28, and the two agree.

In the merge path dependency and licence scan is advisory. Dependency and licence scan costs 4 m 25 s typically and 6 m 50 s when the build fleet is contended. Ownership of dependency and licence scan sits with Security Engineering. dependency and licence scan was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

About secret scan the reviewer would add that introduced after the September 2025 token exposure. Over the reviewed period the secret scan stage improved after the build fleet was enlarged. Duration of secret scan runs 2 m 05 s to 3 m 15 s across the measured window. secret scan carries a follow-up request that was satisfied after the draft of TDR-2026-28 was circulated.

Container image vulnerability scan sits in the pipeline as advisory, and the reviewer verified the configuration rather than the documentation. In lead-time terms container image vulnerability scan contributes the largest single share of any stage. The container image vulnerability scan stage belongs to Security Engineering. container image vulnerability scan appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

Over the reviewed period the performance smoke, cloud stage was unchanged in duration and worse in flakiness. In the merge path performance smoke, cloud is advisory. Flakiness on performance smoke, cloud stands at 3.4% against a total failure rate of 6.6%. performance smoke, cloud was one of two items where TDR-2026-28 could not obtain the underlying data within the engagement.

2.2% of release packaging and signing runs fail and 0.2% of runs fail without a code cause. Trend on release packaging and signing: it was stable in every respect the reviewer could measure. Measured against lead time, release packaging and signing costs a small and predictable share. release packaging and signing is treated as material to the investment case rather than as a routine engineering item.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-28)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
						Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

3. Failure and flake rates, and what they cost the team

In lead-time terms lint and format contributes a small and predictable share. On lint and format the reviewer records that fast and stable; the only stage no engineer complains about. Lint and format is Developer Productivity's to fix when it degrades. lint and format was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Across two quarters unit tests, cloud services degraded after a new rule set was added. The pipeline treats unit tests, cloud services as blocking. The failure rate on unit tests, cloud services is 5.4%, with 1.9% attributable to flakiness. unit tests, cloud services appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

Of the end-to-end lead time, unit tests, robot software accounts for a small and predictable share. Unit tests, robot software is maintained by Autonomy Core. Unit tests, robot software is configured blocking, which is the fact that determines what can reach production. unit tests, robot software appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

Static analysis, robot software is responsible for a share that is invisible because it runs in parallel of the total pipeline time. Ownership of static analysis, robot software sits with Autonomy Core. The reviewer's observation on static analysis, robot software is that the ruleset is narrower than the coding standard requires. static analysis, robot software is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Build, cloud container images was stable in every respect the reviewer could measure across the measured window. Build, cloud container images is Platform Infrastructure's to fix when it degrades. The lead-time cost of build, cloud container images is more than the two preceding stages combined. build, cloud container images was cross-checked against the engineering record supplied for TDR-2026-28, and the two agree.

Reliability of build, robot firmware images is 4.1% failing with 0.6% flaky. Build, robot firmware images is configured blocking, which is the fact that determines what can reach production. Build, robot firmware images contributes more than the two preceding stages combined to the change lead time. build, robot firmware images would move from a concern to a verified defect if one further measurement were supplied.

The reviewer's observation on integration tests, cloud is that the highest flake rate in the pipeline and the most common cause of a retried run. Fleet Services owns integration tests, cloud, and the on-call for pipeline health is a separate rotation. Integration tests, cloud fails on 12.6% of runs, of which 8.2% is flake rather than a real defect. integration tests, cloud is the clearest instance in TDR-2026-28 of a control that exists on paper and is weaker in practice.

Contract tests, warehouse connectors sits in the pipeline as blocking, and the reviewer verified the configuration rather than the documentation. Integrations maintains contract tests, warehouse connectors and answers for its reliability. Contract tests, warehouse connectors is responsible for a share the team regards as acceptable and the reviewer regards as avoidable of the total pipeline time. contract tests, warehouse connectors was checked against the company's own register, which had not been updated since the change landed.

Schedule R - Release history sampled across the reviewed period (TDR-2026-28)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R7.3-101	2025-07-30	Speed and Separation Monitor	full gate set	Farida Cisse	not required	promoted to all rings
R5.11-102	2025-08-04	Fleet Allocator	full gate set	Grace Lindqvist	not required	promoted to all rings
R7.6-103	2025-08-07	Speed and Separation Monitor	full gate set	Grace Lindqvist	not required	promoted to ring one only
R5.12-104	2025-08-11	Map Authoring Studio	full gate set	Priya Raghunathan	rolled back same day	promoted to all rings
R6.7-105	2025-08-15	Config Distribution	emergency path	Priya Raghunathan	not required	withdrawn
R5.2-106	2025-08-21	Order Broker	emergency path	Sonia Krall	rolled back same day	promoted to all rings
R6.4-107	2025-08-25	Fleet Health Monitor	full gate set	Marta Kellerman	rolled back next day	promoted to all rings
R7.1-108	2025-08-27	Order Broker	gate set with the nightly suite waived	Theo Bassett	not required	promoted to all rings
R4.2-109	2025-09-02	Wheel Odometry Filter	gate set with the nightly suite waived	Erik Palmgren	not required	paused at ring two
R5.3-110	2025-09-04	Gripper Controller	gate set with the bench suite waived	Anjali Sekhon	not required	promoted to ring one only
R8.4-111	2025-09-10	Hardware-in-the-Loop Rig	full gate set	Anjali Sekhon	rolled back same day	promoted to ring one only
R6.8-112	2025-09-14	Battery Management Service	full gate set	Ines Duarte	not required	promoted to all rings
R6.2-113	2025-09-18	Fleet Health Monitor	gate set with the nightly suite waived	Bettina Vogel	not required	promoted to all rings
R7.9-114	2025-09-21	Speed and Separation Monitor	gate set with the nightly suite waived	Wanda Njoku	not required	paused at ring two
R5.1-115	2025-09-26	Map Authoring Studio	full gate set	Rahul Menon	not required	promoted to all rings
R5.1-116	2025-09-29	Release Orchestrator	full gate set	Anjali Sekhon	not required	paused at ring two
R8.8-117	2025-10-04	Edge Gateway Agent	gate set with the nightly suite waived	Theo Bassett	not required	promoted to all rings
R7.12-118	2025-10-08	Payload Scale Interface	full gate set	Peter Halvorsen	rolled back same day	withdrawn
R7.10-119	2025-10-12	WMS Connector	gate set with the bench suite waived	Marta Kellerman	not required	promoted to ring one only
R7.11-120	2025-10-15	WMS Connector	full gate set	Sam Oyelaran	not required	withdrawn
R4.10-121	2025-10-18	Robot Shadow Store	full gate set	Ines Duarte	rolled back next day	withdrawn
R7.4-122	2025-10-22	Speed and Separation Monitor	gate set with the nightly suite waived	Bettina Vogel	rolled back same day	paused at ring two
R4.0-123	2025-10-27	Order Broker	full gate set	Sam Oyelaran	rolled back same	withdrawn

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
					day	
R6.7-124	2025-11-01	Safety Controller	full gate set	Yuki Harada	not required	promoted to all rings
R6.10-125	2025-11-05	Hardware-in-the-Loop Rig	full gate set	Wanda Njoku	not required	paused at ring two
R6.3-126	2025-11-08	Alarm Router	full gate set	Wanda Njoku	rolled back same day	paused at ring two
R8.13-127	2025-11-13	Telemetry Ingest	full gate set	Owen Brantley	not required	promoted to all rings
R4.6-128	2025-11-17	Billing Usage Meter	full gate set	Bettina Vogel	not required	promoted to all rings
R5.13-129	2025-11-20	Battery Management Service	gate set with the nightly suite waived	Theo Bassett	not required	promoted to all rings
R7.6-130	2025-11-25	Video Snippet Service	full gate set	Nadia Semenova	rolled back same day	paused at ring two
R5.9-131	2025-11-29	Wheel Odometry Filter	full gate set	Sonia Krall	not required	promoted to all rings
R4.11-132	2025-12-03	Speed and Separation Monitor	full gate set	Colin Achebe	rolled back same day	paused at ring two
R4.12-133	2025-12-07	OTA Updater	full gate set	Owen Brantley	not required	withdrawn
R8.7-134	2025-12-11	Fleet Health Monitor	gate set with the nightly suite waived	Marcus Duplessis	not required	promoted to all rings
R4.13-135	2025-12-15	Robot Shadow Store	gate set with the nightly suite waived	Erik Palmgren	not required	withdrawn
R4.9-136	2025-12-17	Video Snippet Service	gate set with the bench suite waived	Nathan Oduya	not required	promoted to ring one only
R7.3-137	2025-12-22	Motion Planner	full gate set	Wanda Njoku	rolled back same day	withdrawn
R6.10-138	2025-12-26	Fleet Health Monitor	emergency path	Marta Kellerman	not required	withdrawn
R7.2-139	2025-12-31	Simulation Harness	gate set with the bench suite waived	Sam Oyelaran	not required	withdrawn
R4.13-140	2026-01-02	Gripper Controller	full gate set	Colin Achebe	not required	promoted to all rings
R5.0-141	2026-01-06	Edge Gateway Agent	full gate set	Yuki Harada	not required	withdrawn
R6.8-142	2026-01-12	Gripper Controller	full gate set	Marta Kellerman	not required	promoted to all rings
R4.5-143	2026-01-14	Analytics Warehouse Loader	full gate set	Nadia Semenova	not required	promoted to all rings
R4.7-144	2026-01-20	Alarm Router	full gate set	Nathan Oduya	not required	promoted to all rings
R4.2-145	2026-01-24	Hardware-in-the-Loop Rig	full gate set	Anjali Sekhon	not required	promoted to all rings
R6.6-146	2026-01-27	Traffic Manager	gate set with the bench suite waived	Nathan Oduya	rolled back same day	promoted to all rings
R6.4-147	2026-02-01	Gripper Controller	full gate set	Bettina Vogel	rolled back next day	promoted to ring one only
R5.9-148	2026-02-03	Config Distribution	gate set with the bench suite waived	Anjali Sekhon	not required	paused at ring two

3. Failure and flake rates, and what they cost the team - part 2 of 2

Rows of the 3. Failure and flake rates, and what they cost the team resume here, numbered on from the part immediately preceding, so a reference cited once resolves in only one place. The sequence is unbroken either side of it.

Everything stated about how the 3. Failure and flake rates, and what they cost the team was assembled continues to hold, and the break carries no substantive meaning. Nothing is restated from the earlier part.

Schedule R - Release history sampled across the reviewed period (TDR-2026-28) (continued, part 2 of 2)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R7.4-149	2026-02-09	Charge Scheduler	full gate set	Sonia Krall	rolled back same day	paused at ring two
R7.13-150	2026-02-12	Alarm Router	gate set with the nightly suite waived	Nathan Oduya	not required	promoted to all rings
R7.13-151	2026-02-17	Simulation Harness	gate set with the nightly suite waived	Lucia Ferrante	rolled back next day	promoted to all rings
R5.12-152	2026-02-20	E-Stop Chain	gate set with the nightly suite waived	Lucia Ferrante	not required	promoted to all rings
R7.12-153	2026-02-24	Map Authoring Studio	full gate set	Wanda Njoku	rolled back same day	promoted to ring one only
R4.1-154	2026-02-28	WMS Connector	gate set with the nightly suite waived	Ines Duarte	not required	promoted to ring one only
R8.7-155	2026-03-05	OTA Updater	full gate set	Yuki Harada	rolled back next day	withdrawn
R7.8-156	2026-03-08	Telemetry Ingest	gate set with the nightly suite waived	Yuki Harada	rolled back same day	promoted to ring one only
R4.11-157	2026-03-11	Config Distribution	gate set with the nightly suite waived	Marta Kellerman	rolled back next day	promoted to all rings
R4.0-158	2026-03-15	Fleet Allocator	gate set with the nightly suite waived	Sam Oyelaran	rolled back next day	promoted to all rings
R5.9-159	2026-03-21	Alarm Router	emergency path	Sonia Krall	not required	promoted to ring one only
R6.6-160	2026-03-25	Motion Planner	emergency path	Marta Kellerman	rolled back next day	promoted to all rings
R6.10-161	2026-03-28	Alarm Router	gate set with the bench suite waived	Nathan Oduya	not required	paused at ring two
R8.4-162	2026-04-02	Config Distribution	gate set with the nightly suite waived	Farida Cisse	not required	promoted to ring one only
R5.13-163	2026-04-04	Analytics Warehouse Loader	emergency path	Wanda Njoku	not required	promoted to ring one only
R5.4-164	2026-04-10	Charge Scheduler	full gate set	Anjali Sekhon	not required	promoted to ring one only
R5.6-165	2026-04-13	Lidar Driver	full gate set	Farida Cisse	not required	withdrawn
R5.13-166	2026-04-18	Telemetry Ingest	gate set with the bench suite waived	Lucia Ferrante	not required	promoted to all rings
R4.3-167	2026-04-20	Site Provisioning Service	full gate set	Sam Oyelaran	rolled back next day	promoted to all rings
R5.10-168	2026-04-25	Traffic Manager	emergency path	Anjali Sekhon	not required	promoted to all rings
R4.1-169	2026-04-28	Simulation Harness	emergency path	Bettina Vogel	rolled back same day	withdrawn
R7.11-170	2026-05-03	Localization Engine	full gate set	Sam Oyelaran	not required	withdrawn
R7.8-171	2026-05-06	Remote Operations Console	gate set with the nightly suite waived	Grace Lindqvist	rolled back next day	promoted to all rings
R4.7-172	2026-05-12	Lidar Driver	full gate set	Nadia Semenova	not required	promoted to ring one only
R5.12-173	2026-05-16	Video Snippet Service	gate set with the bench suite waived	Nadia Semenova	rolled back next day	withdrawn
R6.12-174	2026-05-19	WMS Connector	full gate set	Bettina Vogel	not required	promoted to all rings
R8.5-175	2026-05-22	Config Distribution	gate set with the nightly suite waived	Anjali Sekhon	not required	paused at ring two
R7.10-176	2026-05-27	Map Authoring Studio	gate set with the nightly suite	Yuki Harada	not required	paused at ring two

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
			waived			
R7.4-177	2026-05-30	Battery Management Service	full gate set	Theo Bassett	rolled back same day	promoted to ring one only
R4.4-178	2026-06-04	Wheel Odometry Filter	gate set with the bench suite waived	Wanda Njoku	not required	promoted to all rings
R5.6-179	2026-06-08	Robot Shadow Store	full gate set	Farida Cisse	not required	promoted to all rings
R4.10-180	2026-06-12	Site Provisioning Service	full gate set	Marcus Duplessis	not required	promoted to all rings
R7.10-181	2026-06-17	Costmap Fusion	gate set with the nightly suite waived	Ines Duarte	rolled back same day	promoted to ring one only
R4.3-182	2026-06-21	Hardware-in-the-Loop Rig	gate set with the bench suite waived	Sam Oyelaran	rolled back same day	withdrawn
R5.11-183	2026-06-23	Costmap Fusion	gate set with the nightly suite waived	Anjali Sekhon	not required	withdrawn
R8.0-184	2026-06-28	Remote Operations Console	gate set with the bench suite waived	Lucia Ferrante	not required	paused at ring two
R7.11-185	2026-07-03	Simulation Harness	gate set with the nightly suite waived	Theo Bassett	rolled back next day	promoted to all rings
R5.5-186	2026-07-07	Map Authoring Studio	full gate set	Bettina Vogel	not required	promoted to ring one only
R8.6-187	2026-07-10	Map Authoring Studio	gate set with the bench suite waived	Anjali Sekhon	rolled back same day	paused at ring two
R4.11-188	2026-07-14	Release Orchestrator	emergency path	Peter Halvorsen	not required	withdrawn
R8.2-189	2026-07-18	Fleet Allocator	emergency path	Yuki Harada	not required	promoted to all rings
R5.10-190	2026-07-21	Edge Gateway Agent	full gate set	Theo Bassett	not required	promoted to ring one only
R6.7-191	2026-07-26	WMS Connector	full gate set	Daniel Okafor	rolled back next day	promoted to all rings
R5.9-192	2026-07-30	Map Authoring Studio	gate set with the bench suite waived	Sam Oyelaran	rolled back same day	promoted to all rings
R5.10-193	2026-08-04	Motion Planner	emergency path	Yuki Harada	not required	withdrawn
R5.9-194	2026-08-06	Lidar Driver	gate set with the bench suite waived	Colin Achebe	rolled back same day	promoted to all rings
R5.0-195	2026-08-11	Motion Planner	full gate set	Grace Lindqvist	rolled back next day	promoted to ring one only
R4.10-196	2026-08-15	Map Authoring Studio	gate set with the bench suite waived	Yuki Harada	rolled back next day	withdrawn

4. Release cadence and ownership across the estate

End-of-life review of Safety Controller found none; the supplier's support commitment runs to 2031. Taking the evidence together, Safety Controller there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Test coverage on Safety Controller is 91 per cent line coverage and the measurement basis was confirmed with Bettina Vogel. Safety Controller is written in C, written to a constrained coding subset, executes on a safety-rated controller module supplied by a third party, and keeps its state in a non-volatile fault log on the controller. The objective recorded against Safety Controller is that a validated protective-stop trigger de-energises the drive inside 120 ms, and it is reported weekly to the engineering leadership. Safety Controller was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Safety Controller owns its own data and exposes it only through a versioned interface, which the reviewer read. The cadence recorded for Safety Controller is twice a year, with full revalidation of the safety case, which matches the observed release history. The backlog for Safety Controller is 3 items deep and contains an acknowledged defect whereby

the fault log wraps after four thousand entries and the oldest records are discarded. Documents and data inspected for Safety Controller were the supplier assessment report, the internal safety case and eight bench trigger tests. The E-stop chain, the speed and separation monitor and the drive current sensors feed Safety Controller; a failure in Safety Controller is felt by the drive power contactors and the robot fault annunciator. Safety Controller is the reason one section of TDR-2026-28 concludes narrowly rather than generally.

The Data and Telemetry team carries Analytics Warehouse Loader on its service inventory, and Farida Cisse answered for it in interview. 20 items sit in the Analytics Warehouse Loader backlog; the entry that bears on the investment case is that a late-arriving telemetry partition is silently excluded rather than reprocessed. Analytics Warehouse Loader commits that the daily load completes before 06:00 local time at every region, and the objective is measured continuously rather than sampled. A columnar warehouse with daily partitions is the system of record for Analytics Warehouse Loader, a Python 3.11 with SQL transformations component running on cloud, a scheduled batch runner. Analytics Warehouse Loader rests on the load schedule, partition completeness checks and two months of load audit records so far as this review is concerned. Analytics Warehouse Loader is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

For Analytics Warehouse Loader the supportability position is one transformation still runs on a deprecated warehouse function. The dependency map places the telemetry ingest path, the order broker and the billing usage meter above Analytics Warehouse Loader and the customer reporting views and the internal metrics layer below it. Analytics Warehouse Loader ships weekly; the reviewer sampled the last twelve releases. In production Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, unchanged since the last architecture review. Analytics Warehouse Loader was checked for shared-database coupling, and the ownership boundary held. Analytics Warehouse Loader was compared with the equivalent position at two other components, and it is neither the best nor the worst.

2 defects remain open against E-Stop Chain; the one the reviewer pressed on is that one pre-production harness revision remains in the field on nine robots at two sites. Evidence examined for E-Stop Chain comprised the wiring diagrams, the change order file and a witnessed actuation test on four units. In production E-Stop Chain is edge, hardwired, independent of the compute module, unchanged since the last architecture review. E-Stop Chain is written in VHDL with discrete relay logic, executes on a safety-rated hardware assembly, no general-purpose operating system, and keeps its state in no persistent store; events are latched to the safety controller. Interface discipline around E-Stop Chain is adequate; the schema is versioned and change is gated on a compatibility check. E-Stop Chain is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The dependency map places the operator stop buttons and the perimeter interlocks above E-Stop Chain and the safety controller and the drive contactors below it. E-Stop Chain's dependency ageing was checked: none; the relay parts are current production. E-Stop Chain is owned by the Safety Engineering team, with Theo Bassett named as the engineering owner of record. For E-Stop Chain the stated target is that drive power is removed within 250 ms of button actuation on every tested unit, expressed against production traffic rather than a benchmark. The reviewer's judgement on E-Stop Chain is that no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. E-Stop Chain was excluded from an earlier draft of TDR-2026-28 and reinstated after the reviewer obtained the supporting record.

Hardware-in-the-Loop Rig is deployed laboratory, two rigs, one of which lags the current sensor configuration. Hardware-in-the-Loop Rig was built in Python 3.11 test harness around production firmware and has not been rewritten; it still runs on bench hardware in the Fremont laboratory against a local results database, backed up weekly. Hardware-in-the-Loop Rig was examined for unsupported components; the rig controller runs an operating system release that left vendor support in 2025. Evidence examined for Hardware-in-the-Loop Rig comprised the bench suite definition, rig configuration inventory and two firmware release records. Robot Firmware holds Hardware-in-the-Loop Rig, and the ownership entry naming Nathan Oduya matches the on-call rotation. Hardware-in-the-Loop Rig carries a follow-up request that was satisfied after the draft of TDR-2026-28 was circulated.

The interface Hardware-in-the-Loop Rig publishes is versioned, and the firmware release decision record were checked for use of any undocumented field. For Hardware-in-the-Loop Rig the stated target is that every firmware candidate runs the full bench suite before ring one, expressed against production traffic rather than a benchmark. On

Hardware-in-the-Loop Rig the conclusion is qualified: no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Hardware-in-the-Loop Rig depends on the firmware candidate artefacts and the bench sensor stimulators and is depended on by the firmware release decision record. Coverage for Hardware-in-the-Loop Rig was 39 per cent line coverage at the review date, and the trend over the reviewed period was down by two points after a large refactor. Hardware-in-the-Loop Rig was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Motion Planner shows 71 per cent line coverage in the pipeline, and the reviewer notes that the untested share includes the error handling. Autonomy Core tracks 14 open defects on Motion Planner, including the acknowledged case where replan oscillation when two robots negotiate the same narrow-aisle pinch point. Autonomy Core publishes one objective for Motion Planner: plan cycle completes inside 40 ms at the 99th percentile. Reading the dependency map from Motion Planner outward gives the costmap fusion pipeline and the localization engine on the input side and the traffic manager and the on-robot drive supervisor on the output side. Responsibility for Motion Planner was confirmed with Priya Raghunathan, whose team, Autonomy Core, also holds the on-call duty. Motion Planner was checked against the corresponding entry in the company's own register, and the register was current.

The manifests place Motion Planner edge, one instance per robot with no cloud fallback. On paper Motion Planner releases every three weeks, promoted ring by ring, and the observed practice did not diverge from that. Hosting for Motion Planner is on-robot Linux with a real-time preemption patch, and its Rust 1.79 code writes through to a local key-value snapshot flushed every ninety seconds. On Motion Planner the conclusion is qualified: the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. The ageing position for Motion Planner is none; the toolchain is pinned to a supported compiler release, and the reviewer checked it against the vendor statements on file. Motion Planner is the clearest instance in TDR-2026-28 of a control that exists on paper and is weaker in practice.

The cadence recorded for Speed and Separation Monitor is twice a year, with revalidation, which matches the observed release history. Deployment of Speed and Separation Monitor is edge, cross-checked against the safety controller's own inputs, which sets the blast radius of any defect in it. Speed and Separation Monitor shows 87 per cent line coverage in the pipeline, and the reviewer notes that the untested share includes the error handling. One coupling concern applies to Speed and Separation Monitor: a rolling in-memory window with fault snapshots to the safety controller is shared with a second service, so a schema change is not a local decision. The backlog for Speed and Separation Monitor is 4 items deep and contains an acknowledged defect whereby the protective distance calculation uses a fixed operator approach speed rather than a measured one. Speed and Separation Monitor is the reason one section of TDR-2026-28 concludes narrowly rather than generally.

Safety Engineering publishes one objective for Speed and Separation Monitor: commanded speed is reduced before the separation distance falls below the protective distance. Reading the dependency map from Speed and Separation Monitor outward gives the costmap fusion pipeline and the safety-rated scanner inputs on the input side and the safety controller and the motion planner speed limit on the output side. What the reviewer had for Speed and Separation Monitor was the separation monitoring design note, the internal safety case and six bench approach tests; the conclusions below do not reach past it. End-of-life review of Speed and Separation Monitor found none; the sensing hardware is current production. Responsibility for Speed and Separation Monitor was confirmed with Bettina Vogel, whose team, Safety Engineering, also holds the on-call duty. Speed and Separation Monitor is reported here in full because a summary of it in an earlier draft of TDR-2026-28 was misread as an assurance.

OTA Updater was weighed on its coverage, its defect backlog and its dependency position, and management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. 8 defects remain open against OTA Updater; the one the reviewer pressed on is that a paused ring resumes automatically after a service restart, which surprised two site teams. Interface discipline around OTA Updater is adequate; the schema is versioned and change is gated on a compatibility check. Release cadence for OTA Updater is every three weeks, following the autonomy release, and the release records bear that out. The ageing position for OTA Updater is none; artefact signing was moved to the current key format in 2025, and the reviewer checked it against the vendor statements on file. OTA Updater is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Coverage for OTA Updater was 76 per cent line coverage at the review date, and the trend over the reviewed period was up by three points. For OTA Updater the stated target is that an update is either fully applied or fully rolled back, with no mixed-version robot, expressed against production traffic rather than a benchmark. OTA Updater has a single named owner, Erik Palmgren of Platform Infrastructure, which is not true of every service reviewed. OTA Updater is written in Rust 1.79 on the robot, Go 1.22 in the cloud, executes on split: an on-robot agent and a cloud release service, and keeps its state in content-addressed object storage for artefacts, PostgreSQL 14 for state. OTA Updater sits split edge and cloud, with staged rings and a per-site pause control, and no alternative placement is currently supported. OTA Updater illustrates why TDR-2026-28 separates the severity of a finding from the confidence behind it.

The Platform Infrastructure team carries Event Bus on its service inventory, and Erik Palmgren answered for it in interview. The open defect backlog for Event Bus holds 4 items, the most consequential being that one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. Event Bus sits cloud, one cluster per region, three availability zones, and no alternative placement is currently supported. Event Bus reports no meaningful automated unit coverage, verification being by witnessed test instead, measured by the same instrumentation the pipeline uses. Every publishing service feed Event Bus; a failure in Event Bus is felt by every consuming service. Event Bus was checked against the company's own register, which had not been updated since the change landed.

Event Bus's dependency ageing was checked: the broker minor release in production leaves the vendor's support window in 2027. The interface Event Bus publishes is versioned, and every consuming service were checked for use of any undocumented field. For Event Bus the reviewer records that the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. On paper Event Bus releases upgraded twice a year, and the observed practice did not diverge from that. Event Bus commits that publish acknowledgement inside 25 ms at p95 and no partition without a leader, and the objective is measured continuously rather than sampled. Event Bus was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule G - Open defect population and ageing by service (TDR-2026-28)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2026-01-10	1 high, 1 medium, 12 low	12	Priya Raghunathan
Costmap Fusion	21	2025-08-29	1 high, 6 medium, 14 low	3	Daniel Okafor
Fleet Allocator	9	2025-09-09	0 high, 3 medium, 6 low	7	Marta Kellerman
Traffic Manager	12	2026-04-02	2 high, 1 medium, 9 low	5	Owen Brantley
Charge Scheduler	7	2025-04-29	0 high, 3 medium, 4 low	13	Nadia Semenova
Safety Controller	3	2025-04-22	1 high, 1 medium, 1 low	7	Bettina Vogel
E-Stop Chain	2	2026-04-11	2 high, 2 medium, 0 low	1	Theo Bassett
Lidar Driver	18	2025-05-10	2 high, 3 medium, 13 low	4	Yuki Harada
Localization Engine	16	2025-12-17	0 high, 4 medium, 12 low	6	Rahul Menon
WMS Connector	23	2025-12-05	0 high, 7 medium, 16 low	6	Tomas Villareal
Order Broker	11	2025-09-04	0 high, 3 medium, 8 low	13	Grace Lindqvist
Telemetry Ingest	19	2025-10-13	2 high, 7 medium, 10 low	12	Farida Cisse
OTA Updater	8	2025-03-02	2 high, 3 medium, 3 low	12	Erik Palmgren
Identity and Authorization	6	2025-06-17	1 high, 2 medium, 3 low	14	Sonia Krall
Remote Operations Console	27	2025-12-16	1 high, 3 medium, 23 low	3	Anjali Sekhon
Simulation Harness	31	2025-08-24	0 high, 3 medium, 28 low	5	Colin Achebe
Hardware-in-the-Loop Rig	13	2025-07-29	1 high, 2 medium, 10 low	9	Nathan Oduya

4. Release cadence and ownership across the estate - part 2 of 2

Presentation of the 4. Release cadence and ownership across the estate carries on below without restarting the row numbers, the break being one of pagination only.

No basis of compilation changes at this break; whatever the 4. Release cadence and ownership across the estate recorded about scope and method at its head continues to apply. The parts are to be read together.

Schedule G - Open defect population and ageing by service (TDR-2026-28) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2025-03-31	1 high, 7 medium, 9 low	13	Wanda Njoku
Battery Management Service	10	2026-03-20	1 high, 2 medium, 7 low	12	Peter Halvorsen
Gripper Controller	15	2025-03-19	0 high, 5 medium, 10 low	2	Ines Duarte
Map Authoring Studio	22	2026-02-15	2 high, 10 medium, 10 low	7	Lucia Ferrante
Site Provisioning Service	5	2025-03-08	2 high, 2 medium, 1 low	9	Marcus Duplessis
Event Bus	4	2025-03-30	1 high, 2 medium, 1 low	8	Erik Palmgren
Config Distribution	6	2025-06-21	1 high, 2 medium, 3 low	7	Sam Oyelaran
Robot Shadow Store	9	2026-03-28	0 high, 2 medium, 7 low	13	Marta Kellerman
Analytics Warehouse Loader	20	2026-04-23	1 high, 10 medium, 9 low	2	Farida Cisse
Billing Usage Meter	7	2026-03-31	0 high, 1 medium, 6 low	12	Nadia Semenova
Alarm Router	14	2025-03-30	2 high, 2 medium, 10 low	5	Wanda Njoku
Video Snippet Service	12	2025-10-17	0 high, 5 medium, 7 low	11	Anjali Sekhon
Speed and Separation Monitor	4	2025-10-17	2 high, 2 medium, 0 low	13	Bettina Vogel
Wheel Odometry Filter	8	2025-08-12	1 high, 4 medium, 3 low	9	Yuki Harada
Payload Scale Interface	11	2025-12-08	1 high, 2 medium, 8 low	9	Ines Duarte
Edge Gateway Agent	5	2025-09-05	2 high, 2 medium, 1 low	4	Sam Oyelaran
Release Orchestrator	16	2025-11-23	1 high, 8 medium, 7 low	12	Colin Achebe

5. Change-related incidents and restore times

Not applicable; no service impact elapsed before incident INC-2025-096 was mitigated, with resolution at 26 days. Incident INC-2025-096 carries limited, being a single account corroborated only by the ticket timeline evidence, and the reviewer declined to raise the finding beyond what that supports. On 2025-08-05 the record for internal incident INC-2025-096 opens: an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. Incident INC-2025-096 traces back to a single fact: the routing table fell back to a mailbox rather than to a secondary responder. The judgement recorded against process gap event INC-2025-096 is that an operational workaround masked the defect for longer than the incident record suggests. INC-2025-096 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

How INC-2025-096 surfaced matters to the assessment: discovered the next morning when the alarm was found unacknowledged. Process gap event INC-2025-096 closure status is the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. In incident INC-2025-096, the rotation source of truth was a spreadsheet maintained outside the alarm router, which is a process finding rather than a code finding. Process gap is the class of Alarm Router incident INC-2025-096; Alarm Router is the component. The blast radius of incident INC-2025-096 was no production site and no robot directly, the exposure being to internal systems. incident INC-2025-096 is treated as material to the investment case rather than as a routine engineering item.

What state persistence event INC-2025-084 tells the reviewer is that the blast radius was set by a deployment topology decision made years earlier. Blue Mesa Components incident INC-2025-084 produced one commitment: to persist the pause state and require a design note for any change to the release state machine, owned by Erik Palmgren, dated 2025-08-15. Grading INC-2025-084 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. Blue Mesa Components incident INC-2025-084 was raised on 2025-07-08 after a paused update ring resumed after a routine service restart and eleven robots took an unintended firmware step. Time to mitigate state persistence incident INC-2025-084 was 27 m, and time to resolve was 5 days. OTA Updater incident INC-2025-084 is the clearest instance in TDR-2026-28 of a control that exists on paper and is weaker in practice.

The contributing factor recorded against state persistence incident INC-2025-084 is that the pause control had been added in a hotfix without a design review. Cause analysis in OTA Updater incident INC-2025-084 concluded that the pause flag was held in process memory rather than in the release state record. Landed on 2025-07-14 is the position on the incident INC-2025-084 commitment as at the review date. Classifying state persistence incident INC-2025-084 as state persistence matters because the OTA Updater backlog contains two further items of the same kind. Scope of impact in incident INC-2025-084 was one site and 11 robots, taken from the fleet records rather than from the narrative. state persistence event INC-2025-084 was cross-checked against the engineering record supplied for TDR-2026-28, and the two agree.

Attention came to internal incident INC-2025-118 when the dependency and secret scanner flagged it on the following scan. Drawing the evidence in secret exposure event INC-2025-118 together, the remediation addresses the immediate cause and leaves the contributing factor in place. Beyond the direct cause, incident INC-2025-118 shows that build logs were retained far longer than the token lifetime, widening the exposure window. Redaction landed on 2025-09-19; retention was changed on 2025-09-24 is the position on the incident INC-2025-118 commitment as at the review date. Mitigation of internal incident INC-2025-118 took 3 h 40 m to revoke and rotate and full resolution took 7 days. secret exposure event INC-2025-118 has no single accountable owner, and that is the finding the reviewer would press hardest.

Identity and Authorization incident INC-2025-118 was contained to no production site and no robot directly, the exposure being to internal systems. For INC-2025-118 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. The trigger recorded for internal incident INC-2025-118 on 2025-09-17 was that a service token with a twelve-hour lifetime was found in a build log retained for ninety days. The classification applied to internal incident INC-2025-118 is secret exposure, and the affected component is Identity and Authorization. Following INC-2025-118 the team undertook to redact authorization headers at the logging layer and cut build log retention to fourteen days, with Sonia Krall accountable by 2025-10-31. Identity and Authorization incident INC-2025-118 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The corrective action for INC-2025-103 is to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, and 2025-10-31 is the date the team gave. In algorithm defect event INC-2025-103 the first signal was the queue depth alert introduced after the March incident fired correctly, and the reviewer treats that as alerted detection. The affected population in Ironwood Grocery Group incident INC-2025-103 was one site and 2 robots. Algorithm defect is the class of algorithm defect incident INC-2025-103; Motion Planner is the component. A contributing factor in algorithm defect incident INC-2025-103 was that the simulation scenario for this geometry existed but had been excluded from the nightly suite. algorithm defect incident INC-2025-103 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Assessed on its own record, INC-2025-103 shows that the failure was in the detection design rather than in the code that failed. Verification of the Ironwood Grocery Group incident INC-2025-103 remediation gives the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. The reviewer grades the algorithm defect incident INC-2025-103 evidence as strong on the cause and limited on the contributing factor and has not upgraded the finding to compensate. Algorithm defect event INC-2025-103 traces back to a single fact: the yield heuristic had no tie-break and both robots computed the same deference score. Reading Ironwood Grocery Group incident INC-2025-103 from its first entry, on 2025-08-22 two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes. algorithm defect incident INC-2025-103 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Cause analysis in incident INC-2026-057 concluded that the estimator was fitted on packs that follow a full charge cycle. Dunbar Industrial felt Dunbar Industrial incident INC-2026-057, which affected 2 sites and 5 robots. Detection came through a robot derated mid-shift and the pack was pulled for analysis, which places modelling event INC-2026-057 on the reported side of the ledger. Following INC-2026-057 the team undertook to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, with Peter Halvorsen accountable by 2026-09-30. The classification applied to Dunbar Industrial incident INC-2026-057 is modelling, and the affected component is Battery Management Service. modelling event INC-2026-057 was checked against the corresponding entry in the company's own register, and the register was current.

Time to mitigate INC-2026-057 was not a service outage; robots derated, and time to resolve was open at the review date. The trigger recorded for Dunbar Industrial incident INC-2026-057 on 2026-05-19 was that state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Not landed; data collection is running and the imbalance alert is in review is the position on the incident INC-2026-057 commitment as at the review date. Dunbar Industrial incident INC-2026-057 carries a second-order finding: the two affected sites moved to opportunity charging after the estimator had been validated. On incident INC-2026-057 the evidence is limited; the log retention window had closed before the reviewer asked for the raw data; the severity assessment is stated separately below. INC-2026-057 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Following incident INC-2026-004 the team undertook to replace the fallback matcher and re-survey the affected aisles, with Rahul Menon accountable by 2026-06-30. INC-2026-004 was raised on 2026-01-13 after relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. Localization Engine accounts for performance event INC-2026-004, a performance event. In INC-2026-004, the site was surveyed with a sparse feature set that increases cache misses, which is a process finding rather than a code finding. The reviewer grades the performance event INC-2026-004 evidence as strong, resting on the production traces the team supplied and has not upgraded the finding to compensate. Kestrel Medical Supply incident INC-2026-004 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The detection path for performance incident INC-2026-004 was the site's commissioning engineer raised it during acceptance testing. Performance event INC-2026-004 took not a service outage; 5 days to a workaround to stabilise and open at the review date to close out. The reviewer's reading of performance incident INC-2026-004 is that the blast radius was set by a deployment topology decision made years earlier. Root cause in incident INC-2026-004 was that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. So far as reach is concerned, INC-2026-004 involved one site and 18 robots. performance incident INC-2026-004 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Drawing the evidence in incident INC-2025-139 together, the remediation addresses the immediate cause and leaves the contributing factor in place. Event Bus owns Event Bus incident INC-2025-139, which the reviewer classes as capacity. The two durations recorded for multiple sites incident INC-2025-139 are 2 h 47 m to mitigate and 63 days to resolve. Event Bus incident INC-2025-139 carries a second-order finding: the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. Grading incident INC-2025-139 on evidence rather than on consequence gives strong, resting on the production traces the team supplied. capacity incident INC-2025-139 was one of two items where TDR-2026-28 could not obtain the underlying data within the engagement.

The trigger recorded for multiple sites incident INC-2025-139 on 2025-11-02 was that a telemetry backlog on a shared topic delayed control events by up to eleven seconds. As to whether the Event Bus incident INC-2025-139 remediation actually landed: the split landed on 2026-01-04; the quarterly review has not yet run. The blast radius of INC-2025-139 was 7 sites and 246 robots. Attention came to capacity incident INC-2025-139 when the traffic manager's own latency alert fired first. Capacity incident INC-2025-139 traces back to a single fact: control and telemetry traffic shared one topic and one consumer group budget. capacity event INC-2025-139 would be a candidate for the first post-close engineering review under any reasonable plan.

Resource leak incident INC-2025-041 took 1 h 52 m to stabilise and 9 days to close out. Traffic Manager incident INC-2025-041 is not read as an isolated failure: the same class of defect can recur in two sibling components that were not changed. INC-2025-041 classifies as a resource leak event against Traffic Manager. Attention came to resource leak incident INC-2025-041 when a site supervisor telephoned the on-call engineer after robots queued at one intersection.

Incident INC-2025-041 affected 2 sites and 61 robots. Traffic Manager incident INC-2025-041 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The contributing factor recorded against resource leak incident INC-2025-041 is that the alert rule for queue depth existed but was scoped to a metric the site did not emit. Owen Brantley committed after resource leak event INC-2025-041 to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, due 2025-05-30. Reading incident INC-2025-041 from its first entry, on 2025-03-14 a robot lost its network link mid-traverse and its corridor reservation was never released. The reviewer checked the resource leak incident INC-2025-041 commitment against the code and release records; landed on 2025-05-22, verified in the reservation replay suite. Root cause in incident INC-2025-041 was that the reservation lease had no expiry, so release depended on the departing robot. resource leak incident INC-2025-041 is reported at the evidence strength available and has not been promoted to fit the narrative.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2026-28)

Drill	Date	Component	Scope	Result	Gap identified
RD-2025-01	2025-08-31	Robot Shadow Store	one region	succeeded with an observation	none
RD-2025-02	2025-09-16	Event Bus	the full fleet in staging	aborted	restore time above target
RD-2025-03	2025-10-03	Billing Usage Meter	one ring	succeeded	none
RD-2025-04	2025-10-17	Fleet Health Monitor	the full fleet in staging	aborted	restore time above target
RD-2025-05	2025-11-03	Traffic Manager	one site	aborted	approval not recorded
RD-2025-06	2025-11-19	Telemetry Ingest	one ring	failed and was retried	state left in the paused ring
RD-2025-07	2025-12-07	Battery Management Service	one region	succeeded	approval not recorded
RD-2025-08	2025-12-22	Safety Controller	one ring	failed and was retried	state left in the paused ring
RD-2025-09	2026-01-08	Payload Scale Interface	one robot	succeeded	approval not recorded
RD-2025-10	2026-01-23	Telemetry Ingest	the full fleet in staging	succeeded	restore time above target
RD-2025-11	2026-02-05	Remote Operations Console	one robot	aborted	restore time above target
RD-2025-12	2026-02-21	E-Stop Chain	one site	succeeded	none
RD-2025-13	2026-03-09	Wheel Odometry Filter	the full fleet in staging	failed and was retried	restore time above target
RD-2025-14	2026-03-28	Fleet Allocator	one robot	succeeded	runbook step missing

5. Change-related incidents and restore times - part 2 of 2

Set out next is the balance of the 5. Change-related incidents and restore times, in the sequence in which the rows were originally compiled. No row is omitted at the join.

Readers should carry the 5. Change-related incidents and restore times's stated conventions across the break, since this part adds no definition of its own. The break is editorial.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2026-28) (continued, part 2 of 2)

Drill	Date	Component	Scope	Result	Gap identified
RD-2026-01	2026-04-14	Release Orchestrator	one region	succeeded	state left in the paused ring
RD-2026-02	2026-04-26	E-Stop Chain	one site	succeeded	state left in the paused ring
RD-2026-03	2026-05-14	Video Snippet Service	one robot	failed and was retried	state left in the paused ring
RD-2026-04	2026-05-28	E-Stop Chain	one robot	aborted	runbook step missing
RD-2026-05	2026-06-17	Lidar Driver	one site	aborted	restore time above target
RD-2026-06	2026-07-01	Traffic Manager	one site	succeeded with an observation	runbook step missing
RD-2026-07	2026-07-19	Billing Usage Meter	one region	succeeded	state left in the paused ring
RD-2026-08	2026-08-03	Hardware-in-the-Loop Rig	one ring	aborted	state left in the paused ring
RD-2026-09	2026-08-18	Costmap Fusion	one robot	failed and was retried	state left in the paused ring

Drill	Date	Component	Scope	Result	Gap identified
RD-2026-10	2026-09-03	Billing Usage Meter	one robot	succeeded with an observation	approval not recorded
RD-2026-11	2026-09-20	Localization Engine	one region	failed and was retried	none
RD-2026-12	2026-10-04	Fleet Health Monitor	one robot	failed and was retried	state left in the paused ring
RD-2026-13	2026-10-19	Site Provisioning Service	one site	succeeded	none
RD-2026-14	2026-11-07	Video Snippet Service	the full fleet in staging	aborted	state left in the paused ring

6. Tooling and test-infrastructure debt

The estimate for register item TD-18 is 0.75 engineer-quarters and $104,000. Under safety evidence, entry programme item TD-18 is to aggregate the safety controller fault log to the cloud continuously. Backlog is the status of TD-18, targeted at 2027-03-31. Safety Controller item TD-18 is one of the entries where TDR-2026-28 relies on the company's own measurement rather than an independent one.

Remote Operations Console item TD-08 cannot start earlier than it does because it is last in a three-item chain, so its date is the least reliable on the register. 1 engineer-quarter is the estimate on security supportability item TD-08, which converts to $132,000. Security supportability item TD-08 carries this observation: no advisory affects the current version, which is why it ranks low. register item TD-08 is the clearest instance in TDR-2026-28 of engineering practice being better than the documentation suggests.

In progress is the status of TD-11 on Site Provisioning Service, targeted at 2026-09-30. The configuration entry at configuration item TD-11 is to re-converge the four forked site provisioning templates. Register item TD-11 appears in the register and is not funded, and carried in the register regardless. TD-11 on Site Provisioning Service rests on a single source, and the reviewer has said so rather than implying corroboration.

For Payload Scale Interface item TD-22 the team estimates 0.25 engineer-quarters, which the reviewer converted to $42,000 for comparison. Funding for TD-22 is funded only if the team's headcount plan is met, and the register does not say so. In the programme TD-22 is ahead of two other items that depend on it. Payload Scale Interface item TD-22 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Costmap Fusion item TD-24 carries this observation: counsel was engaged after the reviewer raised it. What IP hygiene item TD-24 protects against is the fact that it three modified files under a file-level copyleft licence appear unpublished. The plan treats programme item TD-24 as funded in the current financial year, which the reviewer checked against the headcount model. TD-24 on Costmap Fusion was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

TD-14 on Hardware-in-the-Loop Rig carries a 2027-03-31 target and a status of backlog. Sequencing of register item TD-14 matters: it is in parallel with the release-process changes. The delivery entry at delivery item TD-14 is to introduce a second bench rig queue to cut firmware lead time. register item TD-14 would be a candidate for the first post-close engineering review under any reasonable plan.

The register entry Fleet Allocator item TD-01 reads: partition the fleet allocator write path, in the scalability category. Reading scalability item TD-01 against the funded plan, the load-test series identifies this as the single binding scalability constraint. Register item TD-01 is last in a three-item chain, so its date is the least reliable on the register, which is why the target date is later than the effort estimate implies. TD-01 is reported here in full because a summary of it in an earlier draft of TDR-2026-28 was misread as an assurance.

As at the review date TD-17 was in design, with 2026-11-30 still the stated target. Effort on programme item TD-17 is estimated at 0.5 engineer-quarters, costed at $72,000. The reviewer's qualification on TD-17 is that the control is designed but the signing tool does not enforce it. programme item TD-17 was excluded from an earlier draft of TDR-2026-28 and reinstated after the reviewer obtained the supporting record.

Register item TD-03 is on the register because it no upstream fixes for a component on the safety-adjacent path. Under support, entry register item TD-03 is to retire the vendored geometry library in costmap fusion. TD-03 on Costmap Fusion is sequenced behind a platform change that has its own target date. register item TD-03 is included because a reader of TDR-2026-28 should be able to disagree with the inference without repeating the work.

Register item TD-07 carries this observation: the check is written but not enabled as at the review date. Reading the programme plan, tooling item TD-07 is last in a three-item chain, so its date is the least reliable on the register. Effort on TD-07 on Simulation Harness is estimated at 0.5 engineer-quarters, costed at $68,000. Simulation Harness item TD-07 is the entry that changed most between the first and second document requests behind TDR-2026-28.

Reading the programme plan, resilience item TD-12 is behind a supplier release outside the company's control. If resilience item TD-12 is not done, the exposure is that it telemetry loss on outages beyond four hours. For resilience item TD-12 the team estimates 1 engineer-quarter, which the reviewer converted to $144,000 for comparison. register item TD-12 rests on a single source, and the reviewer has said so rather than implying corroboration.

Depends on the next compute module revision is the point the reviewer would put to management on security item TD-16. Security item TD-16 carries an estimate of 1 engineer-quarter; the costed figure is $126,000. What security item TD-16 protects against is the fact that it a stolen compute module discloses customer warehouse layouts. TD-16 on Localization Engine is included because a reader of TDR-2026-28 should be able to disagree with the inference without repeating the work.

Exposure carried by TD-02 is that it strong-copyleft exposure in shipped robot software plus an unsupported dependency. Item Localization Engine item TD-02 in the debt register is to replace the unmaintained scan-matching library in the localization fallback, and it bears on Localization Engine. 2026-12-31 is the target for register item TD-02, whose status is committed. IP and support item TD-02 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

TD-15 is unfunded and dependent on capital rather than engineering effort; the register status and the funding position are not the same thing. For TD-15 on Alarm Router the team estimates 0.25 engineer-quarters, which the reviewer converted to $34,000 for comparison. Register item TD-15 cannot start earlier than it does because it is last in a three-item chain, so its date is the least reliable on the register. security item TD-15 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

7. Findings in the build and release path

The reviewer declines to soften assessment finding SEC-2026-16: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. Costmap Fusion finding SEC-2026-16 was opened on 2026-04-22 and states that costmap fusion diagnostic socket bound to all interfaces. On assessment finding SEC-2026-16 the tester's evidence was that connected from a second host on the robot network, and the reviewer grades it moderate, because the demonstration was performed in a laboratory tenant rather than in production. The tester scored assessment finding SEC-2026-16 at 5.8 and labelled it Medium. Medium-severity finding SEC-2026-16 was raised again with the owning engineer after the initial interview, and the position did not change.

Billing Usage Meter finding SEC-2026-10 rests on the observation that bucket configuration read from the cloud export; the reviewer holds that to be strong. Billing Usage Meter finding SEC-2026-10 produces this judgement: the fix is verified and the class of defect is not, since no test now prevents its return. Severity on Billing Usage Meter finding SEC-2026-10 is Low, scored 4.0 by the assessor. Low-severity finding SEC-2026-10 translates into the following exposure: an export could be altered after the fact. Low-severity finding SEC-2026-10 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Assessment finding SEC-2025-12 translates into the following exposure: a loader defect or compromise could alter source telemetry. 2025-09-25 is the discovery date of SEC-2025-12 against Analytics Warehouse Loader, which finds that analytics warehouse service account held write access to raw telemetry. The reviewer's position on SEC-2025-12 is that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Substantiating SEC-2025-12 is the fact that permissions read from the warehouse grant inventory, graded limited, resting on a configuration read rather than on a demonstrated exploitation. SEC-2025-12 against Analytics Warehouse Loader is reported here in full because a summary of it in an earlier draft of TDR-2026-28 was misread as an assurance.

Finding SEC-2025-04 was graded Medium, band 5.4, by Redwater Security engagement RW-2025-3. After the fix, Map Authoring Studio finding SEC-2025-04 carries residual risk described as medium; the shortened publication path is still outstanding. Assessment finding SEC-2025-04 rests on the observation that reproduced, and independently observed in a field incident; the reviewer holds that to be strong. SEC-2025-04 is remediated, with a closure date of 2025-11-27.

Medium-severity finding SEC-2025-04 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Medium-severity finding SEC-2025-08 produces this judgement: the item is open past its committed date and no revised date has been agreed. SEC-2025-08 reduces to a residual risk of low; the interface is restricted to the operations subnet. Evidence behind finding SEC-2025-08 is that reachability proven by the tester from a deployed container, which the reviewer treats as moderate. Event bus administrative interface reachable from the application subnet is the substance of Medium-severity finding SEC-2025-08, raised on 2025-09-23. Event Bus finding SEC-2025-08 was checked against the company's own register, which had not been updated since the change landed.

SEC-2025-02 matters because any engineer with build log access could replay a service credential. SEC-2025-02 against Release Orchestrator is supported by the fact that the tester extracted a working token from a retained log; evidence strength is limited, resting on a configuration read rather than on a demonstrated exploitation. Assessment finding SEC-2025-02 records that build logs printed the full authorization header, discovered on 2025-09-17. Residual risk after remediation of SEC-2025-02 against Release Orchestrator is low; redaction is enforced in the logging library. Release Orchestrator finding SEC-2025-02 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-28.

As at the review date assessment finding SEC-2026-17 was remediated and the closure entry reads 2026-06-19. Finding SEC-2026-17 sits against Motion Planner and records that motion planner parameter override accepted from an unauthenticated local port. Severity on finding SEC-2026-17 is High, scored 7.9 by the assessor. Finding SEC-2026-17 matters because a local compromise could alter planner limits without detection. assessment finding SEC-2026-17 is one of the entries where TDR-2026-28 relies on the company's own measurement rather than an independent one.

Assessment finding SEC-2025-03 sits against Remote Operations Console and records that remote operations session relay accepted an unpinned certificate chain. Assessment finding SEC-2025-03 stands remediated; 2026-01-22 is what the record gives for closure. The practical consequence in SEC-2025-03 is that an on-path attacker on a customer network could observe operator commands. The reviewer's position on Remote Operations Console finding SEC-2025-03 is that the remediation is adequate and the time to remediate is not. assessment finding SEC-2025-03 was reviewed against the position twelve months earlier, and nothing measurable has changed.

2026-04-20 is the discovery date of finding SEC-2026-13, which finds that order broker accepted an oversized payload without a limit. What SEC-2026-13 enables is this: a large payload degrades broker throughput. Finding SEC-2026-13 stands remediated; 2026-05-15 is what the record gives for closure. Low is the rating finding SEC-2026-13 carries, at 3.3 on the assessor's own scale. Low-severity finding SEC-2026-13 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

SEC-2026-11 arrives as a Low finding with a 4.4 score, and the reviewer did not adjust either. Open is the status of finding SEC-2026-11, dated committed 2026-11-30. The reviewer's position on Low-severity finding SEC-2026-11 is that the fix is verified and the class of defect is not, since no test now prevents its return. On assessment finding SEC-2026-11 the tester's evidence was that file permissions observed on a laboratory robot, and the reviewer grades it moderate. SEC-2026-11 against Gripper Controller was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

8. Performance gates and their calibration

Load run LT-2025-20 came out pass with observation. LT-2025-20 was judged against the internal objective that no telemetry is lost for an outage under four hours. Database failover, allocator primary run LT-2025-20 exists in the series to isolate the size of the concurrent operator session pool. LT-2025-20 appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

Weighed against the projected fleet, 720-robot run LT-2026-01 indicates that the reviewer would repeat this run with the newest site's order profile before relying on it. In six sites after the topic split run LT-2026-01 the bottleneck was the control topic split removed the telemetry coupling. The environment behind scenario LT-2026-01 carries a smaller order history than production and a synthetic warehouse endpoint, and the reviewer weighted the result accordingly. run LT-2026-01 appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

Load run LT-2026-02 was marked fail against the 1,200 ms p95 target by the performance team. The environment behind eight sites, shared region run LT-2026-02 carries the production cloud configuration and a laboratory network profile, and the reviewer weighted the result accordingly. 1,340 ms at p95 is the figure that matters in scenario LT-2026-02, against 455 ms median and 2,880 ms at the tail. load run LT-2026-02 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Comparing load run LT-2026-03 with its neighbour in the series, the difference is the presence of a failure injection. What limited scenario LT-2026-03 was read offload bought headroom without a schema change. LT-2026-03 is not read as a headline number: the reviewer would repeat this run with the newest site's order profile before relying on it. load run LT-2026-03 was raised again with the owning engineer after the initial interview, and the position did not change.

The 1,500 ms p95 acknowledgement objective for the warehouse connector is the target applied to ten sites, shared region run LT-2026-04. The constraint surfaced by ten sites, shared region run LT-2026-04 is the allocator write path needs partitioning to go further. 1200-robot run LT-2026-04 ran against an environment with the current production topology with one region held out. load run LT-2026-04 appears twice in the working papers for TDR-2026-28 because the first answer was corrected after a follow-up request.

What scenario LT-2026-05 does and does not establish is worth stating: the headroom claim is supportable at the tested scale and not at the scale in the operating plan. The order burst after the broker queue change test, recorded as LT-2026-05, was executed on 2026-02-18. Before reading run LT-2026-05, note that the environment carries a synthetic robot fleet driven from recorded mission traces. load run LT-2026-05 is reported at the evidence strength available and has not been promoted to fit the narrative.

Outcome of load run LT-2026-06 was pass with observation, and the reviewer read the raw result rather than the summary. 720-robot run LT-2026-06 is not read as a headline number: the improvement between the two runs is real and its durability is untested under a shard rebalance. Scenario LT-2026-06 used the standard performance environment, which is characterised by the production cloud configuration and a laboratory network profile. run LT-2026-06 is the reason one section of TDR-2026-28 concludes narrowly rather than generally.

The 1,500 ms p95 acknowledgement objective for the warehouse connector is the target applied to scenario LT-2026-07. Reading run LT-2026-07 for the binding constraint gives token issuance was not a limiting term at any tested scale. Set beside the other runs, LT-2026-07 varies only in the presence of a failure injection. scenario LT-2026-07 rests on a single source, and the reviewer has said so rather than implying corroboration.

Console session concurrency run LT-2026-08 supports this conclusion: the improvement between the two runs is real and its durability is untested under a shard rebalance. Load run LT-2026-08 is scored against the fifteen-minute map publication objective, which the company set in its own service documentation. The environment behind load run LT-2026-08 carries a fleet simulator rather than physical robots, with the control loop unchanged, and the reviewer weighted the result accordingly. LT-2026-08 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Weighed against the projected fleet, scenario LT-2026-09 indicates that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. In the series console session concurrency at twice the plan run LT-2026-09 is the run that changes the size of the concurrent operator session pool. 480 robots and 9,400 orders an hour define scenario LT-2026-09, run on 2026-02-25. run LT-2026-09 has no single accountable owner, and that is the finding the reviewer would press hardest.

Run LT-2026-10 supports this conclusion: the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. 240 robots and 8,800 orders an hour define scenario LT-2026-10, run on 2026-02-26. The point of warehouse connector burst, one tenant run LT-2026-10 within the series is whether the read replicas were enabled. scenario LT-2026-10 is the reason one section of TDR-2026-28 concludes narrowly rather than generally.

The warehouse connector burst after outbox partitioning run at warehouse connector burst after outbox partitioning run LT-2026-11 was executed on 2026-04-08 at 240 robots and 8,800 orders an hour. One platform change between otherwise identical runs is what separates warehouse connector burst after outbox partitioning run LT-2026-11 from the adjacent run. Before reading load run LT-2026-11, note that the environment carries a fleet simulator rather than physical

robots, with the control loop unchanged. load run LT-2026-11 was raised again with the owning engineer after the initial interview, and the position did not change.

Pass is what the test record gives for telemetry ingest at twice the current fleet run LT-2026-12. Applying the fifteen-minute map publication objective to 2400-robot run LT-2026-12 produces the verdict recorded above. LT-2026-12 returned 1.4 s to console median, 22 s at the 95th percentile and 48 s at the 99th, erroring at 0.08%. telemetry ingest at twice the current fleet run LT-2026-12 is the clearest instance in TDR-2026-28 of a control that exists on paper and is weaker in practice.

The constraint surfaced by LT-2026-13 is the allocator partitioning programme is the gating dependency. The reviewer's reading of LT-2026-13 is that one run does not establish a trend, and this scenario was executed once. Load run LT-2026-13 exercised the twelve sites, projected 2027 fleet case with 1440 robots and 27,000 orders an hour, on 2026-04-14. load run LT-2026-13 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The four hour edge buffer drain after an outage run at load run LT-2026-14 was executed on 2026-04-15 at 720 robots and 13,800 orders an hour. In 720-robot run LT-2026-14 the median was not applicable and the tail reached not applicable; errors ran at 0.00%. The target LT-2026-14 was measured against is the fifteen-minute map publication objective. 720-robot run LT-2026-14 rests on a single source, and the reviewer has said so rather than implying corroboration.

9. How a release reaches the sites

Bethlehem, Pennsylvania is sparse survey features, which is why relocalisation is slow here, and the reviewer weighted the sample accordingly. Site ST-14 is Bethlehem, Pennsylvania: Kestrel Medical Supply, 37 robots, live from 2025-11. Bethlehem, Pennsylvania was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Edwardsville, Illinois came online in 2026-03 and now carries 24 robots for Crestline Fulfillment. Against its peak profile, Edwardsville, Illinois retains 63% headroom on a vendor B integration. Edwardsville, Illinois was raised again with the owning engineer after the initial interview, and the position did not change.

Against its peak profile, Lehigh Valley, Pennsylvania retains 38% headroom on a vendor B integration. 64 robots are deployed at Lehigh Valley, Pennsylvania, which went live in 2024-02. Lehigh Valley, Pennsylvania is the clearest instance in TDR-2026-28 of engineering practice being better than the documentation suggests.

For Chino, California the material fact is that it is the go-live delayed by the forked provisioning template. Chino, California runs 33 robots for Larkspur Home and opened in 2026-01. Chino, California was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Kenosha, Wisconsin runs 41 robots for Dunbar Industrial and opened in 2024-11. 49% of headroom remains at Kenosha, Wisconsin, whose connector targets vendor D. Kenosha, Wisconsin was one of two items where TDR-2026-28 could not obtain the underlying data within the engagement.

For Whitestown, Indiana the material fact is that it is the site with the standing map publication workaround. Remaining capacity at Whitestown, Indiana is 45%, measured against the site's peak order profile. Whitestown, Indiana was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Remaining capacity at Reno, Nevada is 57%, measured against the site's peak order profile. Reno, Nevada is the smallest customer deployment, and the reviewer weighted the sample accordingly. Reno, Nevada is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

At Groveport, Ohio the fleet is 52 robots, and the site has been live since 2024-09. Groveport, Ohio is notable in this review because it is opportunity charging only, which is where the pack estimator drift appeared. Groveport, Ohio was inspected in the repository rather than described from the ticket, which is why the account is specific.

Headroom at Tracy, California stands at 36%; vendor B is the integrated warehouse system. The Tracy, California deployment serves Harborline Systems with 58 robots, live since 2025-01. Tracy, California was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Newnan, Georgia is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall, which is why the reviewer sampled it. The Newnan, Georgia deployment serves Ironwood Grocery Group with 66 robots, live since 2025-07. Newnan, Georgia is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The Mesquite, Texas site integrates with vendor C and shows 71% headroom. Mesquite, Texas came online in 2026-06 and now carries 16 robots for Evergreen Warehousing. Mesquite, Texas is the clearest instance in TDR-2026-28 of a control that exists on paper and is weaker in practice.

Wilmington, Ohio runs 112 robots for Crestline Fulfillment and opened in 2023-08. On Wilmington, Ohio the record shows that it is the largest deployment and the closest to the allocator scaling limit. Wilmington, Ohio would move from a concern to a verified defect if one further measurement were supplied.

Vendor A is the warehouse system at Lockbourne, Ohio, where headroom is 68%. Site ST-17 is Lockbourne, Ohio: Granite Peak Distribution, 19 robots, live from 2026-04. Lockbourne, Ohio illustrates why TDR-2026-28 separates the severity of a finding from the confidence behind it.

Fathom Mobility operates 47 Helioscope robots at Shepherdsville, Kentucky, commissioned in 2025-03. Shepherdsville, Kentucky was included in the sample because it is manipulator-equipped robots are concentrated here. Shepherdsville, Kentucky is reported here in full because a summary of it in an earlier draft of TDR-2026-28 was misread as an assurance.

On Ontario, California the record shows that it is the site whose shift pattern change caused the April 2025 charge incident. At Ontario, California the fleet is 88 robots, and the site has been live since 2024-05. Ontario, California was checked against the company's own register, which had not been updated since the change landed.

10. Process findings management does not accept

The reviewer asked in each case whether a declined item had a compensating control, and this report reports the answer even where it is unsatisfactory. The report at TDR-2026-28 records 3 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. TDR-2026-28 sampled 9 services from an inventory of 34, weighted towards those on the critical control path. Sampling in TDR-2026-28 covered 9 of the 34 catalogued services and 3 of the 18 live sites. This report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Open items in the present report are stated with what would close each one, so a post-close plan can be costed against them. The open items in the report at TDR-2026-28 are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. D. Achterberg could not verify one class of claim in the present report, namely the behaviour of components deployed only at the two sites not visited. D. Achterberg did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. The present report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

Report TDR-2026-28 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Where the present report says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Two document requests remained partly unsatisfied when report TDR-2026-28 was delivered, and the affected conclusions are marked. TDR-2026-28 is limited by log retention: the telemetry window is ninety days, so events before 2025-09-01 could not be examined directly. Report TDR-2026-28 is one input to a diligence process and is not a substitute for the company's own engineering judgement.

D. Achterberg would add one qualitative item to the pack proposed in TDR-2026-28: the count of commitments closed without verifying evidence. Post-close monitoring under this report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. The approach D. Achterberg took in the report at TDR-2026-28 was to test the company's own assertions rather than to restate them. Report TDR-2026-28 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. Pinehurst Systems Assurance has costed remediation at the company's own loaded engineering rate rather than at a market rate.

11. Conclusions on delivery capability

On the evidence available by 2026-06-18, the report at TDR-2026-28 concludes that management's own view of its technical risk is broadly accurate, with two material understatements identified above. Whatever weight a reader gives the individual findings, TDR-2026-28 concludes that management's own view of its technical risk is broadly accurate, with two material understatements identified above. The reviewer found the interview evidence for the present report candid, and in two cases more candid than the written record. Minutes of every the report at TDR-2026-28 interview were returned to the participant, and three came back with substantive corrections. Pinehurst Systems Assurance was given read access without restriction on scope within the agreed subject matter.

For the transaction, the report at TDR-2026-28 identifies 2 matters the reviewer would raise before closing and 5 for the post-close engineering plan. TDR-2026-28 separates matters that bear on the closing conditions from matters that belong in the first two quarters after close. Pinehurst Systems Assurance delivered this report on 2026-06-18; the scope agreed with Ardent Peak was the engineering delivery process, pipeline metrics and change-management controls. The report at TDR-2026-28 was prepared by Pinehurst Systems Assurance for Ardent Peak and covers the engineering delivery process, pipeline metrics and change-management controls at Helioscope Robotics, Inc. as at 2026-06-18. D. Achterberg interviewed 9 people for the report at TDR-2026-28; where an account was uncorroborated it is marked as such.

The monitoring pack the present report proposes would have surfaced two of the incidents examined here before a customer noticed. Pinehurst Systems Assurance recommends that the 6 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. D. Achterberg was granted read access to the code, the pipeline and the incident tracker for the purposes of the report at TDR-2026-28; production access was not requested. D. Achterberg led the work behind this report, combining document review, 9 interviews and hands-on inspection of running systems. TDR-2026-28 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

Where Wheel Odometry Filter is concerned, the commitment is that odometry drift stays under 1.5 per cent of distance travelled on a clean floor; the reviewer traced the metric to its source. Wheel Odometry Filter has a single named owner, Yuki Harada of Robot Firmware, which is not true of every service reviewed. Wheel Odometry Filter carries 8 open defects, among them a case where drift exceeds the specification on wet or dusty floors and is not flagged to the operator. Wheel Odometry Filter publishes a stable interface, though two fields used by the localization engine and the motion planner are documented only in code comments. Wheel Odometry Filter is promoted with the firmware, twice a year, with the ring order fixed by policy. Wheel Odometry Filter was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

31 defects remain open against Simulation Harness; the one the reviewer pressed on is that scenario determinism breaks when the physics core is rebuilt with a different optimisation level. Simulation Harness publishes a stable interface, though two fields used by the release orchestrator gate and the autonomy regression report are documented only in code comments. Supportability of Simulation Harness turns on one fact: the physics core depends on a library branch that stopped receiving fixes in 2023. Documents and data inspected for Simulation Harness were the scenario catalogue, two weeks of nightly run records and a determinism experiment. Simulation Harness was weighed on its coverage, its defect backlog and its dependency position, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Simulation Harness is not material on its own and belongs in this report because of what it reveals about the surrounding process.